Everyone Can Achieve

SUCTION-EFFECTIVE MANDIBULAR COMPLETE DENTURE

SEMCD



Abe Dental Clinic

Jiro Abe



INTERACTION

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Preface

Many dental researchers are currently embracing the Visual Analog Scale (VAS) and Oral Health Impact Profile(OHIP) to evaluate patient denture satisfaction levels. It is because these methods are useful to create a visible measure to scale patients' level of satisfaction. Nevertheless, these results are concluded based on a series of questions answered by patients but not clinicians. In other words, the results vary between patients and are also very subjective because each patient has an individual level of satisfaction.

To determine the critical components of patient satisfaction level in the complete denture treatments, I propose a new standard on how to evaluate the quality of edentulous impressions by clinicians. Some may argue that the majority of clinicians can quickly assess the success of conventional denture methods, such as the compound impression technique since it has been used and taught in dental schools worldwide. However, this impression technique and its result can be commented on and evaluated differently by various clinicians. Some people may say, *"You took very nice impressions, and the patient will be pleased" others say, "Your impressions are too big. It will cause your patient to be dissatisfied".*

To create a single standard for evaluating an impression of an edentulous mandible, I will first apply a very straightforward method to assess the quality of an edentulous impression of the maxilla.

When you evaluate the quality of maxillary impression for an edentulous patient, the majority of clinicians check the amount of suction by moving the maxillary impression forward and downward to test the creation of negative pressure, which creates a suction seal. Thus, one can expect that the amount of suction effect of the maxillary edentulous impression will be equivalent to the amount of suction when delivering the completed denture prosthesis to the patient. We can usually determine the level of patient denture satisfaction by merely observing the amount of suction at the time of the final impression. By applying the same evaluation methods, we can also determine the quality of an edentulous mandibular impression. When you follow the SEMCD process (Suction-Effective Mandibular Complete Den-

ture) and remove the mandibular edentulous impression from the patient's mouth, you will hear a suction-releasing "Pop" which you have never heard before. Even though some clinicians may have experienced this suction "Pop" sound before they learned the SEMCD method, they probably could not explain why the lower suction sound was created and how to replicate it with other denture patients. Many Clinicians around the world have taken SEMCD hands-on training courses and are applying this technique in their daily practice. It is thus straightforward and effective to use the suction "Pop" sound as the primary measure to evaluate lower denture success.

Many SEMCD lecture attendees have expressed very similar comments and often share their experiences, such as *"Hearing the suction "Pop" sound, made me very happy and I got goosebumps"* or, *"I cannot wait to apply this technique in my practice to create more patient satisfaction"*. It is evident that when you apply simply the suction "Pop" sound as the main key element to evaluate the quality of the edentulous impression of the mandible, you can confidently measure the quality of your impressions and present your advanced skill level to your patients. The SEMCD method allows us to thrive not only as dental professionals but also as members of a worldwide SEMCD family.

Jiro Abe



Foreword

Over the past 18 years, I have experienced successes and frustrations while working with fully-edentulous patients. Making a stable and retentive non-implant supported lower denture has been truly challenging as many patients struggle with loose mandibular dentures. The McGill Consensus Statement on Overdentures in 2002 recommended that "*a 2-implant overdenture should become the first-choice treatment for the edentulous mandible,*" yet very few denture wearers proceed with implant therapy.

My denturist career was transformed in 2014 with the introduction to Dr. Jiro Abe and his SEMCD/lower suction denture technique by Masato Takeuchi, one of my students while I was teaching denturism at Vancouver Community College. He shared Dr. Abe's lower suction denture YouTube video and textbook, Mandibular Suction-Effective Denture and BPS: A Complete Guide, and I was intrigued. With Masato's help, I attempted to implement Dr. Abe's technique and began to achieve lower suction denture success, which was impossible before. Not only was I excited to achieve mandibular suction repeatedly, but my patients reported much-improved denture stability and greater denture satisfaction. SEMCD worked, I needed to learn more and so attended Dr. Abe's introductory SEMCD course in 2015, and returned three more times to Tokyo to become a clinical and technical SEMCD world instructor.

It has been an honour to get to know Dr. Abe's and review his updated BPS/SEMCD textbook. He has devoted his dental career to improve the lives of denture wearers, and this publication contains an easy-to-follow method to achieve this goal. The Bio-functional Prosthetic System (BPS), with it's proven clinical and laboratory materials and tools, is the foundation for denture success. Dr. Abe explains in great detail the essential clinical and technical steps on how to create a suction-effective seal between the mandibular denture borders and oral mucosa. Achieving mandibular suction is no longer a mystery or "simple luck" but is achievable for all levels of denture care providers. SEMCD has transformed my professional career, and I earnestly hope that it will become the new standard to create better-fitting dentures and will be shared in dental educational facilities around the globe. Denture wearers deserve better dentures, and SEMCD is a proven technique that delivers custom-crafted prosthesis.

Thank you, Dr. Abe, for your passion and dedication to improve removable prosthetics and for equipping us with expanded knowledge and new skills to help our edentulous patients.

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August 2020



Foreword

For most of my 27 year career in the dental industry, the world of removable dental prosthetics has often been frowned upon, not seen as a desirable or even considered gold standard treatment . At one juncture, it was suggested to me that nobody would be wearing complete dentures by the year 2000 yet, 20 years later, complete denture treatments account for almost 20% of the total treatment carried out at my clinic in the Republic of Ireland .

Dr. Jiro Abe has devoted a large part of his entire life to the development of (SEMCD) suction mechanism for complete mandibular dentures not only for his own patients but to spreading his pioneering techniques worldwide. This book is an absolute must for all members of a dental team involved in removable dental prosthetics from the most experienced prosthodontist to the young dentist fresh out of university to experienced denturists and dental technicians of all ages and ability. We are all part of a dental team with one goal in mind, providing the highest possible standard of care to our patients.

I first encountered the name Dr. Jiro Abe in 2015 while receiving IvoBase technical training. During this time, I was shown a short video clip of Dr. Abe trying, with difficulty, to remove a complete mandibular denture due to the incredible suction action. I was fascinated and had to seek out Dr. Abe for myself.

I first travelled to Japan in January 2016 for a basic 2 day hands on course and I've since become of or Dr. Abe's trusted lieutenants and disciples of SEMCD. This book is a wonderful collaboration between Dr. Abe's suction mechanism and the BPS system developed by the global dental giant, ivoclar vivadent. This book has been written in a very clear step by step approach which is very easy to understand both clinically and technically. It will inspire anyone who reads it to improve their prosthodontic skills, as it inspired me. Thank you Dr. Abe.

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Introduction

1 What is Suction?

Suction can be defined as "the production of a partial vacuum by reason of reduced air pressure in order to procure adhesion" (Oxford Languages Dictionary).

The author describes lower denture suction as a "negative pressure/suction effect generated when the patient occludes from the mandibular rest position. Saliva under the denture base is discharged at that biting moment generating negative pressure by sealing the entire denture border all around" (**Fig A**).



Fig A Lower denture suction is created by sealing the entire border with oral lining mucosa.

2 How the mandibular denture suction is created?

Let us look at a suction cup to understand how to create negative pressure and adhesion. The internal surface of a suction cup engages a smooth surface. When a force is applied in the center, negative pressure is achieved, which causes a suction effect. If one lifts the cup's periphery, the suction is released, and the cup becomes loose.

It means that the entire denture border seal is mandatory in order to achieve mandibular denture suction.

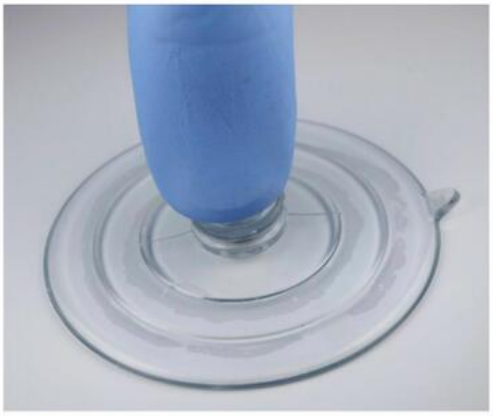


Fig B A suction cup.

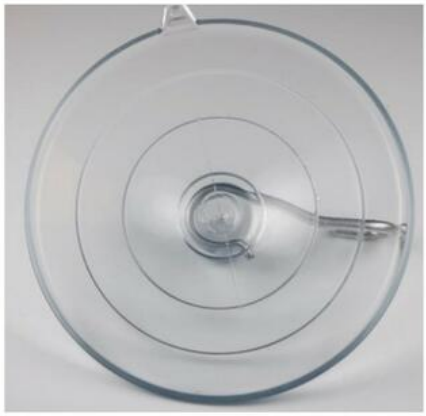


Fig C Suction is activated.

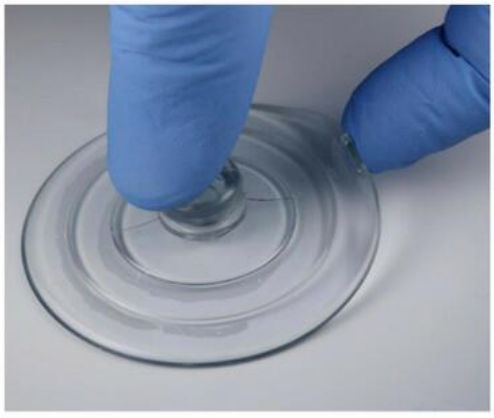


Fig D Suction is released.

(These pictures are provided by Mr. Markus Fischer)

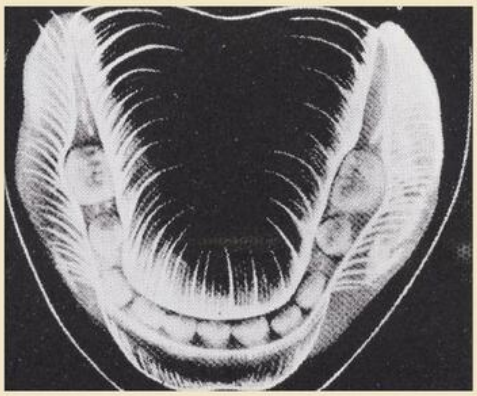
3 Absolute requirements for mandibular denture suction

What do we need to do to achieve an effective suction for the mandibular complete denture? When a hole is drilled in the palatal area of a maxillary denture with a round bur, the denture drops quickly. The maxillary denture is held in place when the oral mucosa seals all its borders, and a tiny air-leaking spot is enough to break this seal. The same principle applies to the mandibular denture. The requirement for mandibular denture suction is precisely the same as for maxillary denture suction, which is to seal the entire peripheral denture borders. It comprises these essential principles (**Figs E, F**).

What is the most important requirement for achieving mandibular denture suction?

↓

Seal all borders!



(Brill N. Tryde G and Contor R (1965))

Fig E Mandibular denture suction is achieved when all borders are sealed with the mobile oral mucosa.

Requirements for mandibular denture	Intraoral status
1. Sealing all denture borders It is essential to seal the entire denture borders with the mobile oral mucosa in order to ensure an effective suction of the mandibular complete denture.	**1. During swallowing (teeth in occlusion)** In order to swallow, posterior teeth must be in contact. The average person is said to swallow about 2000 times a day. Saliva is squeezed out with occlusal pressure during swallowing. 
2. Establishment of suction When occlusal forces are applied to the denture, saliva is squeezed out, temporarily creating a negative pressure to the interior surface of the denture base. The oral mucosa and tongue also provide support.	**2. In mandibular rest position** Maxillary and mandibular denture teeth are out of occlusion when the mandible is at rest. 
3. Maintenance of suction Once suction is established, the denture stays in place when the patient opens the mouth.	**3. During mouth opening** The mobile mucosa exerts lifting force, but suction can be maintained even without occlusal pressure.  

Fig F Mechanism of suction.

4 Happiness brought by SEMCD

When we construct a set of complete dentures for a fully edentulous patient, it is always the mandibular jaw that gives rise to challenges. It is no exaggeration to say that the success of mandibular complete denture fabrication automatically leads to success in treating fully edentulous jaws. Perhaps the most frustrating moment for any dentist is when the finished mandibular denture must be adjusted because it lifts up every time the patient opens the mouth (**Fig G**).

Conversely, when the mandibular denture is held in place by suction, occlusal adjustments are simple to make, which further improves the function of the denture and increases the patient's and clinician's levels of satisfaction. This gives the clinician a strong sense of accomplishment (**Figs H, I**).

The mandibular complete denture lifts on mouth opening.
When the mandibular denture keeps lifting, the patient is discouraged from having dinner or chatting with friends.

No!!



Fig Ga The patient opens her mouth gingerly to prevent the mandibular denture from lifting.



Fig Gb She dared to open the mouth wide.



Fig Gc The mandibular denture lifted as expected. She has been constantly troubled by the denture that keeps lifting up.

Yes!!



Fig Ha Pretreatment: Initial visit.





Fig Hb Posttreatment: Smiling face in full of happiness.

Yes!!







Fig Ia,b Once the mandibular complete denture is effective in suction, both patient and operator are in great comfort.